Separable Differential Equations

ANSWER KEY

Section 1: General Solutions of Separable Equations

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$.

a. $y^2 = 2x^2 + C$ ✓

b. $y = x^2 + C$

c. $y^2 = x^2 + C$

d. $y = 2x^2 + C$

Separate variables: $y\,dy = 2x\,dx$. Integrate both sides: $\frac{1}{2}y^2 = x^2 + C_1 \Rightarrow y^2 = 2x^2 + C$.

2 Find the general solution to $\frac{dy}{dx} = 3x^2 y$.

a. $y = Ce^{x^3}$ ✓

b. $y = e^{x^3} + C$

c. $y = Cx^3$

d. $y = 3x^3 + C$

$\frac{1}{y}\,dy = 3x^2\,dx \Rightarrow \ln|y| = x^3 + C_1 \Rightarrow |y| = e^{C_1}e^{x^3} \Rightarrow y = Ce^{x^3}$.

3 Find the general solution of $\frac{dy}{dx} = (1 + y^2)e^x$.

a. $\arctan(y) = e^x + C$ ✓

b. $y = \tan(e^x) + C$

c. $y = e^x + C$

d. $\ln(1 + y^2) = e^x + C$

$\frac{1}{1+y^2}\,dy = e^x\,dx \Rightarrow \arctan(y) = e^x + C$.

Section 2: Initial Value Problems (IVPs)

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Solve the initial value problem $\frac{dy}{dx} = 2xy$ with $y(0) = 3$.

a. $y = 3e^{x^2}$ ✓

b. $y = e^{x^2} + 2$

c. $y = 3e^{2x}$

d. $y = 3x^2 + 3$

$\frac{dy}{y} = 2x\,dx \Rightarrow \ln|y| = x^2 + C_1 \Rightarrow y = Ce^{x^2}$. Using $y(0) = 3 \Rightarrow C = 3$. Thus $y = 3e^{x^2}$.

5 Solve $\frac{dy}{dx} = -\frac{x}{y}$ with $y(3) = 4$.

a. $x^2 + y^2 = 25$ ✓

b. $x^2 - y^2 = -7$

c. $y = -x + 7$

d. $x^2 + y^2 = 7$

$y\,dy = -x\,dx \Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C_1 \Rightarrow x^2 + y^2 = C$. At $(3, 4)$: $9 + 16 = 25 \Rightarrow x^2 + y^2 = 25$.

6 Solve $\frac{dy}{dt} = ky$ given $y(0) = 50$ and $y(2) = 200$. What is the formula for $y(t)$?

a. $y(t) = 50 \cdot 2^t$ ✓

b. $y(t) = 50e^{2t}$

c. $y(t) = 50 \cdot 4^t$

d. $y(t) = 50 + 75t$

$y(t) = y_0 e^{kt} = 50e^{kt}$. $y(2) = 50e^{2k} = 200 \Rightarrow e^{2k} = 4 \Rightarrow e^k = 2$. Thus $y(t) = 50(2^t)$.

Section 3: Slope Fields and Solution Curves

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Along which curve in the xy-plane do the line segments in the slope field for $\frac{dy}{dx} = x - y$ have slope 0?

a. **Along the line** $y = x$ ✓

b. Along the x-axis ($y = 0$)

c. Along the y-axis ($x = 0$)

d. Along $y = -x$

Slope $= 0 \Rightarrow \frac{dy}{dx} = 0 \Rightarrow x - y = 0 \Rightarrow y = x$.

8 A slope field shows line segments that depend ONLY on y (horizontal rows of equal slopes). Which differential equation could this represent?

a. $\frac{dy}{dx} = y(2 - y)$ ✓

b. $\frac{dy}{dx} = x^2 - y$

c. $\frac{dy}{dx} = xy$

d. $\frac{dy}{dx} = 3x + 1$

If slopes are identical for every x at a fixed y, $\frac{dy}{dx}$ is independent of x, as in autonomous equations like $\frac{dy}{dx} = y(2 - y)$.

9 For $\frac{dy}{dx} = x^2 + y^2$, what can be said about all non-origin slopes in the slope field?

a. **All slopes are strictly positive** ✓

b. Slopes are positive in quadrants I and III only

c. Slopes are 0 everywhere

d. Slopes are negative when $x < 0$

Since $x^2 + y^2 > 0$ for all $(x, y) \neq (0, 0)$, all tangent line segments have strictly positive slope.

Section 4: Logistic Growth Models

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In the logistic differential equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$, what is the carrying capacity M?

a. 800 ✓

b. 0.05

c. 400

d. 16000

The standard logistic equation is $\frac{dP}{dt} = kP(1 - P/M)$, where M is the carrying capacity. Here $M = 800$.

11 At what population P is the growth rate $\frac{dP}{dt}$ maximized in the logistic equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$?

a. $P = 400$ ✓

b. $P = 800$

c. $P = 200$

d. $P = 0$

The growth rate is a quadratic function of P that attains its maximum at half the carrying capacity: $P = M/2 = 800/2 = 400$.

12 What is $\lim_{t \to \infty} P(t)$ for any solution to $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$ with $P(0) = 50$?

a. 800 ✓

b. ∞

c. 400

d. 50

For any initial population $P_0 > 0$, the logistic curve asymptotically approaches the stable carrying capacity $M = 800$ as $t \to \infty$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647